

packages/graph-view/src/utils/layout.worker.ts

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Overview

A new worker module `packages/graph-view/src/utils/layout.worker.ts` is added to perform graph layout calculations using the `@dagrejs/dagre` library. It defines the public contract through `LayoutMessage` (lines 3-7) and `LayoutResult` (lines 10-12) and exposes a message handler that receives node/edge data, runs Dagre, and posts back node positions.

Key changes

- **Import** `dagre` (R1).
- **Exported interfaces** `LayoutMessage` (R3-R7) and `LayoutResult` (R10-R12).
- **Message handler** (`self.onmessage` , R15-R47) parses `LayoutMessage` , builds a Dagre graph, configures layout options (`rankdir` , `nodesep` , etc.), runs `dagre.layout` , and constructs a `positions` map.
- **Result posting** (`self.postMessage` , R46) sends back a `LayoutResult` object, using `satisfies LayoutResult` for type safety.
- The worker is self-contained; no external state is referenced.

Impact

- **Deterministic layout:** given the same node dimensions and edge list, the worker produces the same positions.
- **Performance:** layout is performed in a Web Worker, preventing UI thread blocking for large graphs.
- **Explicit API:** exported interfaces make the worker's contract clear and testable.
- **Dependency:** requires `@dagrejs/dagre` ; ensure it is listed in `package.json` and bundled correctly.

Risks & follow-ups

- **Dependency version:** verify that the installed `@dagrejs/dagre` version matches the API used (`graphLib.Graph`).
- **Worker registration:** confirm that consuming code correctly instantiates this worker and handles its `onmessage` responses.
- **Edge cases:** test with empty node/edge arrays to ensure no runtime errors.
- **Type safety:** ensure that the `satisfies LayoutResult` check passes in all build configurations; adjust TS settings if necessary.